International Cybersecurity and Digital Forensics Academy (ICDFA)

Basic Computer Skills for Digital Forensics

Lab 1B: PC System Fundamentals, Disk Sectors and Partition Analysis

Student Practical Laboratory Manual

Course Code	CIP-B101
Course Title	Basic Computer Skills for Digital Forensics
Lab Code	Lab 1B
Lab Title	PC System Fundamentals, Disk Sectors and Partition Analysis
Estimated Time	3-4 hours
Mode	Individual practical lab with screenshots and written calculations
Required Output	A structured lab report submitted in PDF format
Source Material	Computer System Introduction slide deck

Important instruction before starting

Students must carefully go through the Computer System Introduction slide material before attempting this lab. The slide is a walkthrough of the required concepts: computer system layers, evidence sources, process/thread basics, disk sectors, partition size calculation, PC boot process, MBR/GPT, and file systems. Do not begin the practical section until you understand these concepts.

1. Lab Overview

This lab introduces the basic computer system knowledge required for beginner digital forensic work. Students will identify system hardware and software information, map where digital evidence can exist, calculate disk and partition sizes from sector information, inspect partition layout using safe read-only commands, explain the boot process, and identify file systems used by common operating systems.

Forensic mindset
The goal is not to change anything on the machine. The goal is to observe, record, calculate, verify, and document findings in a repeatable way.

2. Learning Outcomes

- Explain how computer system layers generate digital evidence.
- Identify basic PC hardware, operating system, storage, memory, and processor information.
- Differentiate between a program, process, and thread in a forensic context.
- Explain why computer forensics is technically difficult due to operating systems, applications, hardware, connected devices, and data volume.
- Describe disk sectors and calculate disk or partition sizes using sector count and sector size.
- Use safe read-only commands to view disk and partition information.
- Explain the PC boot process from power-on to operating system control.
- Compare MBR and GPT partitioning concepts at beginner level.
- Identify common file systems including FAT, NTFS, ext, XFS, JFS, btrfs, and HFS+.
- Prepare a professional evidence-style lab report with screenshots, calculations, and interpretations.

3. Safety and Evidence Handling Rules

- Do not format, delete, resize, encrypt, or create partitions during this lab.
- Do not run commands that write changes to disk unless the instruction explicitly says so. This lab is observation and analysis only.
- Use your own computer or approved training VM only.
- If using a shared laboratory machine, obtain instructor permission before collecting screenshots.
- When using Disk Management, DiskPart, PowerShell, WinHex, fdisk, or parted, use them only to view/read information.
- Blur or hide personal serial numbers, email addresses, and usernames if submitting publicly. Do not hide partition sizes or file system evidence required for grading.

4. Required Tools and Environment

Item	Required?	Purpose	Notes
Windows 10/11 PC or Windows VM	Yes	System specification, partition and file system inspection	Use built-in tools such as System Information, Disk Management, CMD, PowerShell, and DiskPart.
Linux/Kali VM	Optional	Alternative partition/file system inspection	Use lsblk, fdisk, parted, df, and blkid as read-only inspection tools.
Screenshot tool	Yes	Evidence documentation	Windows Snipping Tool, Flameshot, GNOME Screenshot, or similar.
Calculator or Python	Yes	Partition and disk size calculation	Students must show formula and

Item	Required?	Purpose	Notes
			calculation steps.
WinHex or approved hex/forensic viewer	Optional	MBR/GPT sector viewing	Only open devices/images in read-only mode. Do not edit sectors.

5. Relevant URLs Extracted from the Slide

The following reference links appeared in the course slide and may help students understand the concepts. They are not a substitute for the practical tasks.

- Technology layer cake - users, apps, OS and hardware: https://windsongtraining.ca/the-technology-layer-cake-users-apps-os-and-hardware/
- Carnegie Cyber Academy - parts of the computer: http://www.carnegiecyberacademy.com/facultyPages/computer/computers.html
- Wikipedia - process in computing: https://en.wikipedia.org/wiki/Process_(computing)
- Hard drive image reference: https://regmedia.co.uk/2006/12/12/fujitsu_300gb_1.jpg
- Disk sector diagram: https://i.imgur.com/26BalHW.png
- Partition diagram reference: https://www.maketecheasier.com/assets/uploads/2012/05/partitions-partition-diagram.png
- PC booting reference: http://www.c-jump.com/CIS24/Slides/Booting/Booting.html
- MBR diagram reference: https://i2.wp.com/neosmart.net/wiki/wp-content/uploads/sites/5/2015/01/Master-Boot-Record.png?resize=1024%2C416&ssl=1
- GPT diagram reference: https://commons.wikimedia.org/w/index.php?curid=3036588
- File systems reference: https://www.geeksforgeeks.org/file-systems-in-operating-system/

6. Lab Scenario

You are a junior digital forensic trainee asked to prepare an initial system profile for a computer that may later be used in a forensic investigation. Your task is to document the system safely, identify possible evidence layers, inspect disk and partition details, calculate partition sizes, explain the boot process, and identify file systems without altering the computer.

Part A - System Identification and Evidence Layer Mapping

In this section, you will identify the major computer system layers and explain how each layer can produce forensic evidence.

A1. Capture basic system specification

- On Windows, open System Information by pressing Windows + R, typing msinfo32, and pressing Enter.
- Capture a screenshot showing OS Name, Version, System Type, Processor, BIOS Mode, Installed Physical Memory, and Secure Boot State if available.
- Also run the command below in Command Prompt and capture the first page of output.

```
systeminfo
```

A2. Capture storage information

- Open Command Prompt or PowerShell.
- Run one of the command sets below. Capture screenshots of storage model, size, partitions, and file systems.

- If a command is not supported on your system, use the alternative command.

```
wmic diskdrive get model,size,serialnumber
wmic logicaldisk get caption,filesystem,size,freespace,volumename

PowerShell alternative:
Get-Disk
Get-Partition
Get-Volume
```

A3. Map evidence sources to system layers

- Create a table in your report showing at least five system layers/sources and the type of evidence that can be generated from each one.
- Use this guide: User activities = Applications + Operating System + Disk/Memory.
- Explain why evidence may exist in more than one layer.

Layer / Source	Example Evidence	Where It May Be Found	Why It Matters
User layer	Opened files, searched terms, typed usernames	Recent files, browser history, shell history	Links user action to system activity
Application layer	Office document metadata, browser cache, chat logs	Application folders and databases	Shows what software created or used evidence
OS layer	Event logs, installed programs, user profiles	Windows logs, registry, Linux /var/log	Shows system events and account activity
Disk layer	Deleted files, file fragments, partitions	File system and unallocated space	Supports recovery and timeline analysis
Memory/process layer	Running malware, PIDs, network connections	RAM capture, process list	Shows live system behavior

Part B - Program, Process and Thread Investigation

This section links the slide concept of program, process, and thread to forensic observation.

B1. Observe running processes

- Open Task Manager on Windows, or use Command Prompt/PowerShell.
- Capture a screenshot showing running processes.
- Select three processes and record their process names, PID values, and likely purpose.

```
tasklist

PowerShell alternative:
Get-Process | Select-Object -First 20 Name,Id,CPU,WorkingSet
```

B2. Explain the difference

- In your report, explain the difference between a program, a process, and a thread in your own words.
- Explain why process information may be useful during malware analysis or live forensic triage.

Part C - Disk Sector and Disk Size Calculation

Key formula

Disk or partition size = number of sectors x bytes per sector. To convert bytes to GB, divide by 1024 x 1024 x 1024.

C1. Reproduce the slide calculation

Use the values from the slide to calculate the total disk size: sectors per track = 400, heads = 12, cylinders = 17000, and bytes per sector = 512.

Item	Formula / Value	Answer
Total sectors	400 x 12 x 17000	81,600,000 sectors
Total bytes	81,600,000 x 512	41,779,200,000 bytes
Kilobytes	41,779,200,000 / 1024	40,800,000 KB
Megabytes	40,800,000 / 1024	39,843.75 MB
Gigabytes	39,843.75 / 1024	38.91 GB

C2. Calculate your own disk or partition size

- Use your partition table or disk details to identify at least two partitions.
- For each partition, record the start sector, number of sectors, sector size, and file system where available.
- Calculate the size of each partition in GB using the formula above.
- Your calculation must be shown step-by-step. Screenshots alone are not enough.

```
Example calculation:
Partition size in GB = sector_count x 512 / 1024 / 1024 / 1024
If sector_count = 83,881,984:
83,881,984 x 512 / 1024 / 1024 / 1024 = 39.99 GB
```

Part D - Safe Partition Table Inspection

Choose either Windows or Linux. You may do both for extra practice. You must not modify the disk.

D1. Windows method - Disk Management, PowerShell and DiskPart

- Open Disk Management by pressing Windows + R, typing diskmgmt.msc, and pressing Enter.
- Capture a screenshot showing your disk layout and visible partitions.
- Open PowerShell as normal user or administrator if required and run the safe read-only commands below.
- Open Command Prompt as administrator and run DiskPart only for listing disks and partitions. Do not run create, delete, clean, format, active, or convert commands.

```
PowerShell:
Get-Disk
Get-Partition
Get-Volume

DiskPart safe listing only:
diskpart
list disk
select disk 0
list partition
exit
```

D2. Linux/Kali method - read-only inspection

- Open Terminal in your Linux/Kali VM or host.

- Run the commands below and capture screenshots showing disk, partition, size, and file system information.
- Do not run mkfs, fdisk write commands, parted write commands, dd write commands, or mount read-write commands for this lab.

```
lsblk -o NAME,SIZE,TYPE,FSTYPE,MOUNTPOINT
sudo fdisk -l
sudo parted -l
blkid
df -h
```

D3. Optional WinHex or forensic viewer method

- If your instructor permits, open the physical disk or disk image in a forensic/hex viewer in read-only mode.
- Navigate to sector 0 and capture the first sector where the MBR/protective MBR may be visible.
- Identify whether the disk appears to use MBR or GPT. If you cannot safely view this, skip the optional step.

Part E - MBR, GPT and Boot Process Explanation

E1. Explain the boot process

- Create a simple flow diagram or numbered list explaining how a PC boots.
- Your explanation must include power-on, POST, BIOS/UEFI, CMOS/firmware settings, MBR/GPT or EFI boot records, boot loader/manager, kernel loading, and operating system control.
- State whether your own system uses BIOS/Legacy or UEFI based on system information evidence.

Stage	What Happens	Forensic Relevance
Power-on and POST	Hardware is initialized and checked.	Hardware failure or tampering may affect evidence access.
BIOS/UEFI firmware	Boot device order and firmware settings are applied.	Boot mode can explain partition style and boot artifacts.
MBR/GPT or EFI boot	System locates bootable partition or EFI boot entry.	Boot records may contain evidence of boot configuration.
Boot loader/manager	OS selection and pre-kernel loading occurs.	May show multi-boot or hidden OS behavior.
Kernel and OS control	Operating system fully starts.	Logs and user activity begin to accumulate.

E2. Compare MBR and GPT

Feature	MBR	GPT
Age / design	Older partitioning system from the 1980s.	Modern partitioning system used with UEFI systems.
Size limitation	Commonly limited to about 2 TB.	Supports very large disks; slide notes up to about 9.4 ZB.
Partition information	Partition table is stored in the first sector with boot code.	Uses primary and backup GPT headers and partition entries.
Forensic issue	Critical information in sector 0; damage can affect partition interpretation.	Backup structures may help recovery and validation.

Part F - File System Identification and Interpretation

- Identify the file system used by at least two volumes or partitions on your system.
- For Windows systems, identify whether volumes use NTFS, FAT/FAT32, exFAT, or another file system.
- For Linux systems, identify whether partitions use ext4, XFS, btrfs, swap, or another file system.
- In your report, explain why a file system is important in digital forensics.

```
Windows:
wmic logicaldisk get caption,filesystem,size,freespace,volumename
Get-Volume

Linux:
lsblk -f
blkid
df -Th
```

File System	Common OS / Usage	Forensic Importance
FAT/FAT32	Removable media and older systems	Simple structure; often used on USB drives.
NTFS	Modern Windows systems	Includes metadata such as MFT entries, timestamps, permissions, and journals.
ext family	Linux systems	Uses inode-based metadata and supports forensic timeline analysis.
XFS/JFS/btrfs	Linux/Unix-like systems	Advanced features may affect recovery and metadata interpretation.
HFS+	Older macOS systems	Important in Mac forensic investigations; slide notes date support limitations.

Part G - Final Practical Assignment

Main assignment

Verify the size of partitions 1 and 2 of a PC using a partition table. Record the verification process step-by-step. Calculate the size of each partition in GB. No full credit will be awarded if you do not use partition table evidence.

Task	Required Evidence	Student Notes / Answer
Identify partition 1	Screenshot of partition table and command output	
Record partition 1 sector count	Start sector, end sector or sector count	
Calculate partition 1 size	Formula and result in GB	
Identify partition 2	Screenshot of partition table and command output	
Record partition 2 sector count	Start sector, end sector or sector count	
Calculate partition 2 size	Formula and result in GB	
Compare with OS-reported size	Disk Management, Get-Volume, lsblk, or df output	
Explain difference if any	Short explanation of rounding, reserved space, file system overhead, or GiB/GB differences	

8. Student Report Template

- Cover page: student name, registration number, course code, lab title, date, and instructor name.
- Introduction: state the purpose of the lab in 5-8 lines.
- Tools used: list operating system, commands, and any software used.
- System specification evidence: include screenshots and explain important observations.
- Evidence layer mapping: include the required table.
- Process observation: include process/PID evidence and explanation.
- Disk and partition inspection: include partition screenshots and command output.
- Manual calculations: show formulas and results for at least two partitions.
- Boot process explanation: include your diagram or numbered boot sequence.
- File system identification: explain the file systems found.
- Conclusion: summarize what you learned and why this matters in digital forensics.
- Appendix: include extra screenshots if necessary.

9. Screenshot Checklist

No.	Screenshot Required	Included?
1	System Information or systeminfo output	Yes / No
2	Disk drive model and size output	Yes / No
3	Logical volume/file system output	Yes / No
4	Running process/PID output	Yes / No
5	Disk Management, Get-Partition, DiskPart, lsblk, fdisk, or parted output	Yes / No
6	Partition 1 calculation evidence	Yes / No
7	Partition 2 calculation evidence	Yes / No
8	Boot mode evidence: BIOS/UEFI	Yes / No
9	File system identification evidence	Yes / No
10	Final report summary/conclusion page	Yes / No

10. Marking Rubric - 100 Marks

Assessment Area	Marks	What Instructor Will Check
Preparation and safety compliance	10	Student followed read-only approach and did not perform destructive actions.
System specification and evidence layer mapping	15	Clear screenshots and explanation of system layers and evidence sources.
Program/process/thread explanation	10	Correct process evidence and explanation of forensic relevance.
Disk sector and size calculation	20	Accurate formulas, calculations, unit conversions, and explanation.
Partition table inspection	20	Reliable partition evidence from approved commands/tools; two partitions analyzed.

Assessment Area	Marks	What Instructor Will Check
Boot process and MBR/GPT explanation	10	Correct boot sequence and comparison of MBR/GPT.
File system identification	10	Correct file system evidence and interpretation.
Report quality and screenshots	5	Professional formatting, clear screenshots, and complete submission.

11. Common Mistakes to Avoid

- Submitting screenshots without manual calculation steps.
- Using only total disk size and failing to identify partitions 1 and 2.
- Confusing GB and GiB rounding differences without explanation.
- Running destructive disk commands such as clean, format, delete, mkfs, or dd to a physical disk.
- Failing to explain why file systems matter in digital forensics.
- Copying slide definitions without adding your own observations and screenshots.

12. Source Acknowledgement

This lab was prepared from the course slide material titled Computer System Introduction for CIP-B101 Basic Computer Skills for Digital Forensics. Students are expected to read the slide material first because it provides the walkthrough for the practical work.